An Institute of

NIE NATIONAL INSTITUTE OF EDUCATION SINGAPORE

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

SINGAPORE MATHEMATICS CURRICULUM

Sun Daojun

3 February 2026

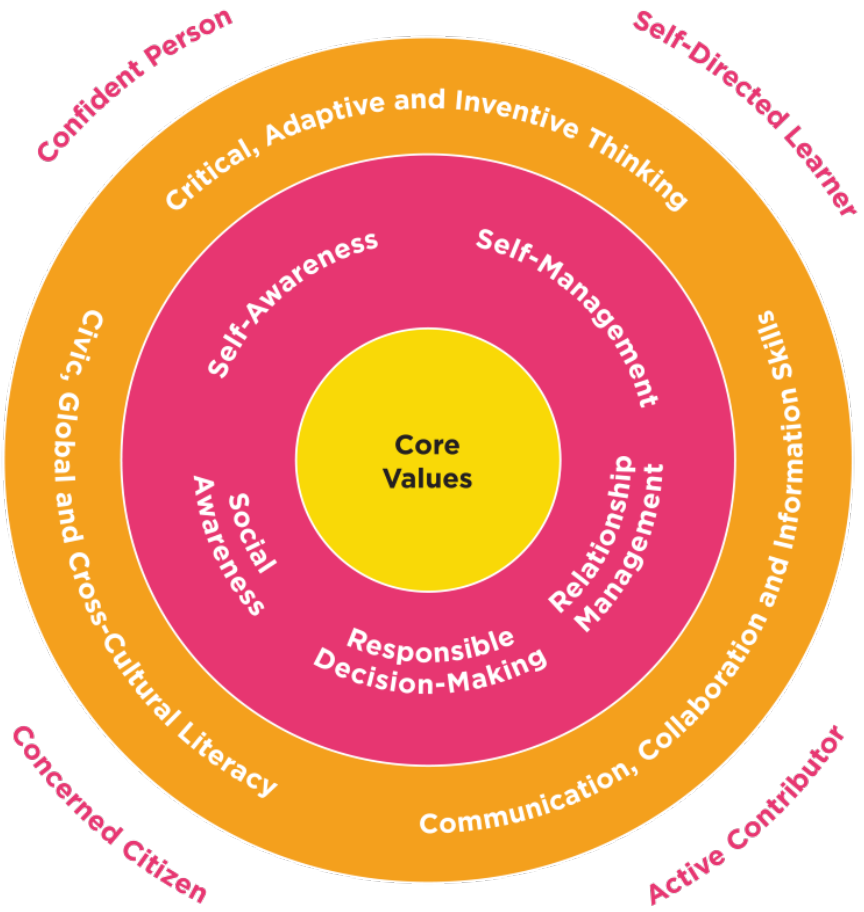
INSPIRING LEARNING
TRANSFORMING TEACHING
ADVANCING RESEARCH

Overview of Lecture

- Background
- Curriculum Design
- Singapore Secondary Mathematics Curriculum
- Problem Solving

Background

Globalisation, changing demographics and technological advancements are some of the key driving forces of our current times, and they will continue to shape our future. Our students will have to be prepared to face these challenges and seize new and exciting opportunities. (MOE, 2023)



©2023 Ministry of Education, Singapore

Curriculum

What is A (Math) Curriculum?
(Padlet - Section 1A)

Curriculum Types
(Padlet - Section 1B)

https://tinyurl.com/ssmc2026

Curriculum

What is A (Math) Curriculum?

- A set of instructional materials
- Syllabuses
- Scheme of work
- Everything taught both inside and outside of school that is directed by the school (Oliva & Gordon, 2013)

At the school level

Curriculum Types

Intended Curriculum

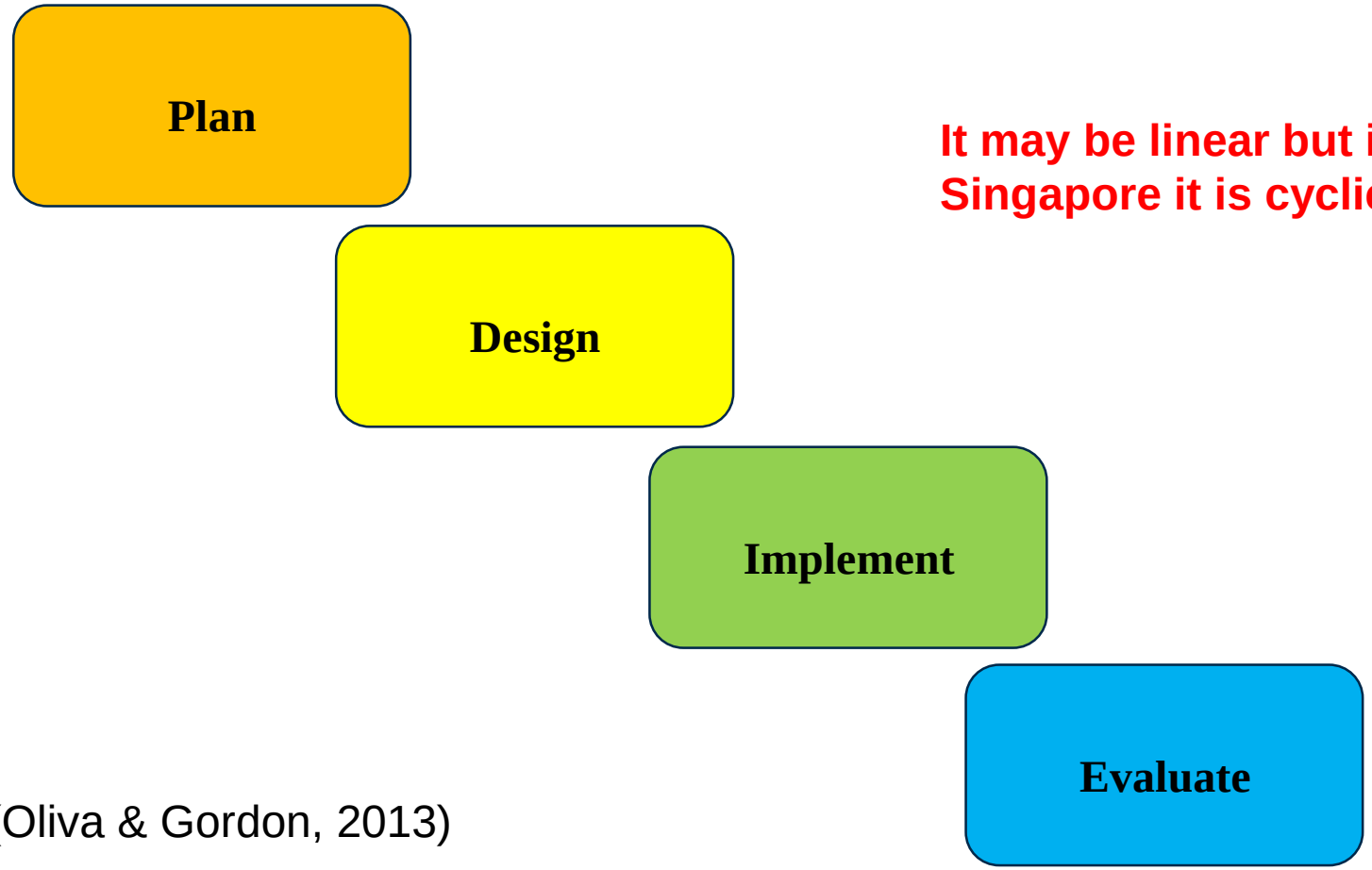
Planned Curriculum

Enacted Curriculum

Experienced Curriculum

Curriculum

Phases of the Development of a Curriculum



It may be linear but in Singapore it is cyclic.

(Oliva & Gordon, 2013)

Singapore Secondary Mathematics Syllabuses

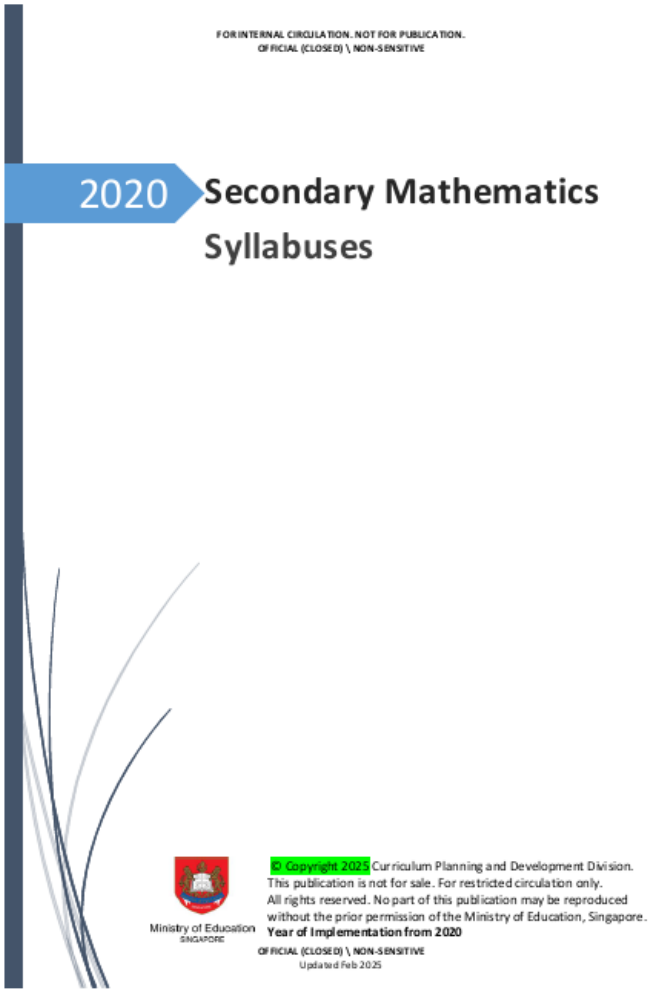
Section 1: Introduction

Section 2: Mathematics Curriculum

Section 3: Mathematics Syllabuses

Section 4: Teaching, Learning and Assessing

Section 5: Summative Assessment


FOR INTERNAL CIRCULATION. NOT FOR PUBLICATION.
OFFICIAL (CLOSED) \ NON-SENSITIVE

2020 Secondary Mathematics Syllabuses

Ministry of Education SINGAPORE

© Copyright 2025 Curriculum Planning and Development Division.
This publication is not for sale. For restricted circulation only.
All rights reserved. No part of this publication may be reproduced without the prior permission of the Ministry of Education, Singapore.
Year of Implementation from 2020
OFFICIAL (CLOSED) \ NON-SENSITIVE
Updated Feb 2025

Singapore Secondary Mathematics Syllabuses

Goals

- To ensure that all students will achieve a level of **mastery** of mathematics that will enable them to function effectively in **everyday life**; and
- For those who have the interest and the ability, to learn more mathematics so that they can **pursue** mathematics or mathematics related courses of study in the **next stage of education**.

Orientations

- **Learning mathematics as a tool** – emphasis on using mathematics as a tool to solve problems where students likely to apply mathematics;
- **Learning mathematics as a discipline** – emphasis on understanding the nature of mathematics, illustrating the thinking and practices of mathematicians.

Aims

- To learn and to apply **concepts** and **skills** to solve problems, including those in contexts;
- To develop **process** and **metacognitive** skills through a mathematical approach to problem solving;
- To inculcate **positive attitudes** towards mathematics

Goals, Aims and Orientations

- What does it mean to teachers?
- Understanding the goals, aims and orientations of syllabus
  - helps teachers **stay focused** on the larger outcomes of learning and
  - guides teachers when they embark on school-based curriculum **innovation and customisation**

Key Features of the Curriculum

- **Spiral** curriculum

  Mathematics is largely hierarchical in nature . Higher concepts and skills are built upon the more fundamental ones and have to be learned in sequence a spiral approach is adopted in the building up of content across the levels.

- **Connected** Syllabuses

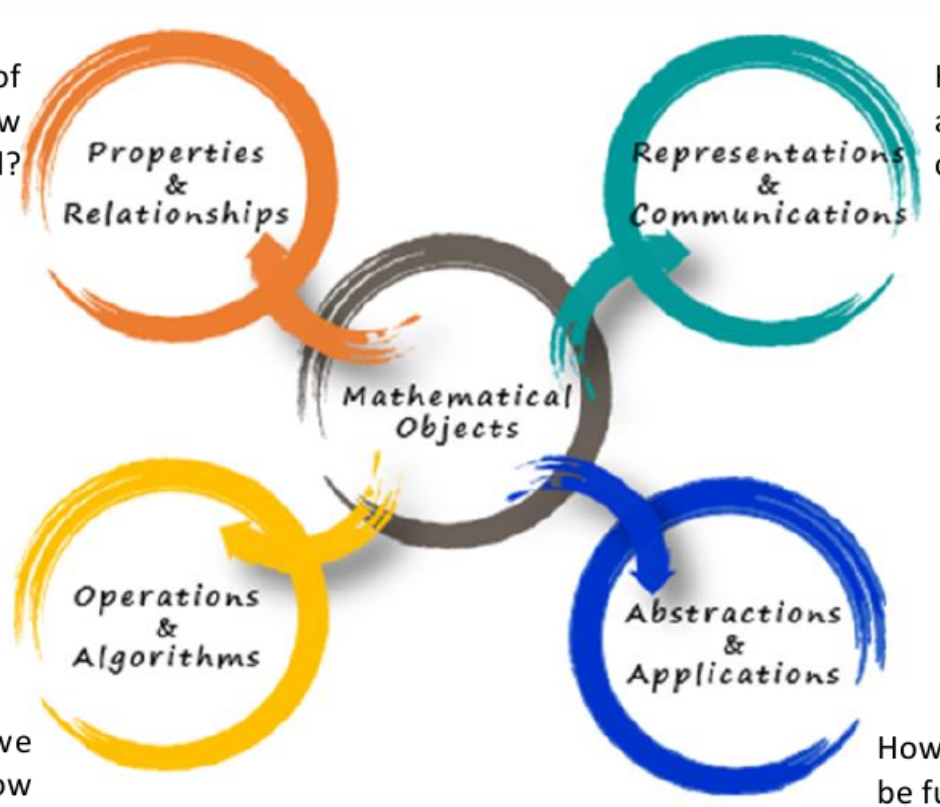
  The mathematics curriculum consists of a set of connected syllabuses (G1/G2/G3) to cater to the different needs and abilities of students.

Nature of Mathematics
- Themes

- There are 4 recurring themes in the study of mathematics.

What are the properties of mathematical objects and how are they related?

How can the mathematical objects and concepts be represented and communicated?



What meaningful actions can we perform on the objects and how do we carry them out?

How can the mathematical objects be further abstracted and where can they be applied?

Nature of Mathematics - Big Ideas

- Big ideas express ideas that are central to mathematics. They appear in different topics and strands.
- There is a continuation of the ideas across levels.
- They bring coherence and show connections across different topics, strands and levels.
- Eight (8) clusters of big ideas are listed in this syllabus.

Functions	**Measures**
Invariance	**Equivalence**
Notations	**Proportionality**
Diagrams	**Models**

Your Top 3 Big Ideas (Section 2A)

https://tinyurl.com/ssmc2026

Nature of Mathematics - Big Ideas

GAI give the same top 3 and provide similar reasons.

- Top 3 Big Ideas (in my view)

Proportionality

It connects many major topics.

Proportional reasoning underpins:
- Fractions, ratios, percentages
- Rates and speed
- Similarity and scale drawing
- Probability
- Measurement (conversion of units)
- Linear functions etc

A student who struggles with proportionality often struggles *everywhere else*, even if procedures are taught well.

Equivalence

It is about sameness across representations.

Examples:
- Numbers:
  6 / 7 = 12 / 14;
  1 / 2 = 0.5
- Algebra:
  2(x+3) = 2x+6
  Use equivalent transformations to solve equations
  Four different representations of a function, etc

Functions

It studies relationships.

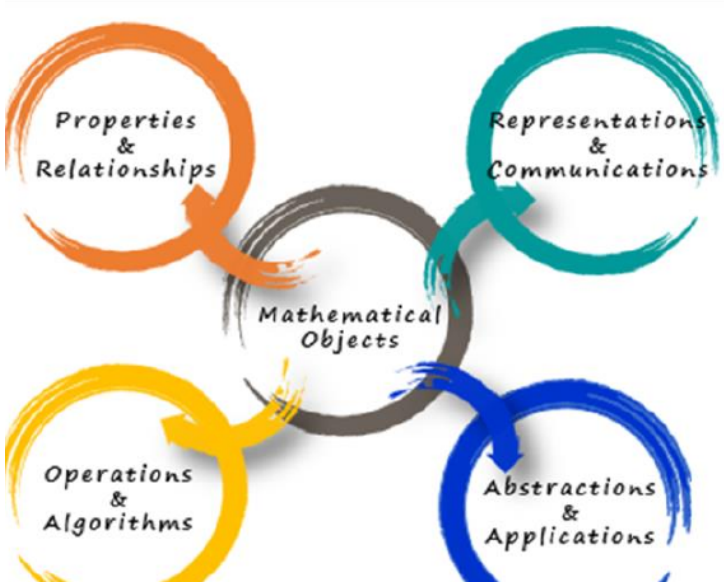
•They make sense of change

Functions describe and predict how one quantity affects another.

•They bridge mathematics and science

Kinematics, growth, decay, and trends all depend on functional reasoning.

Nature of Mathematics
- Connection between themes and big ideas



Section 2B

https://tinyurl.com/ssmc2026

Functions	**Measures**
Invariance	**Equivalence**
Notations	**Proportionality**
Diagrams	**Models**

Nature of Mathematics
- Connection between themes and big ideas

Equi valence

Func tions

Invariance

Proportionality

Notations

Diagrams

Func tions

Measures

Models

Equi valence

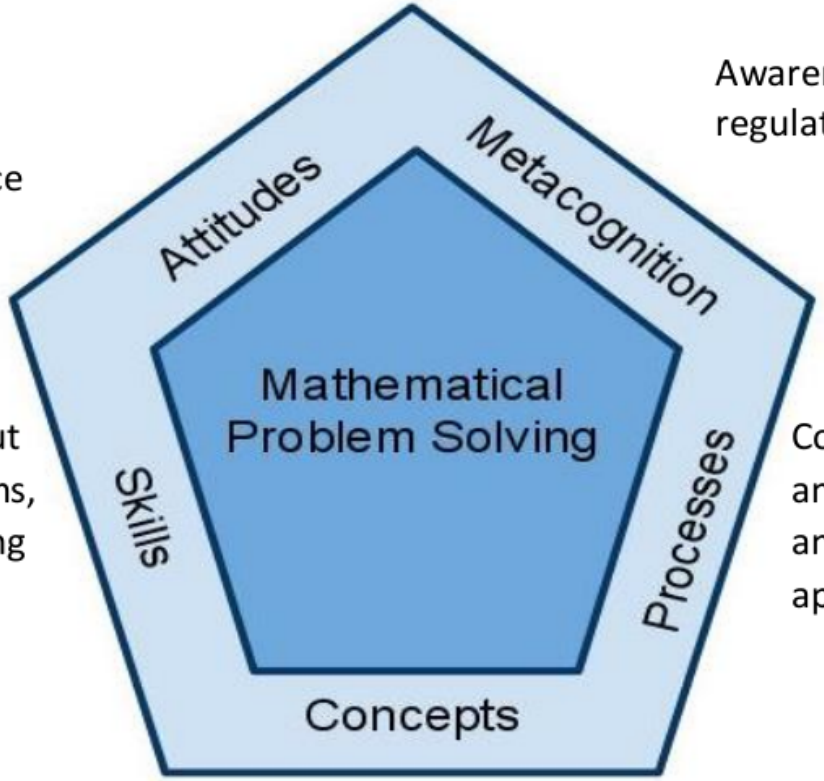
Themes and Big Ideas

- What does it mean to teachers?
- The four (4) themes and eight (8) big ideas are integral parts of the Singapore secondary math curriculum. Teachers must be cognizant of the themes and big ideas so as to enact the curriculum comprehensively.

Singapore Mathematics Framework
- The Current Pentagon

Belief, appreciation, confidence, motivation, interest and perseverance

Awareness, monitoring and regulation of thought processes

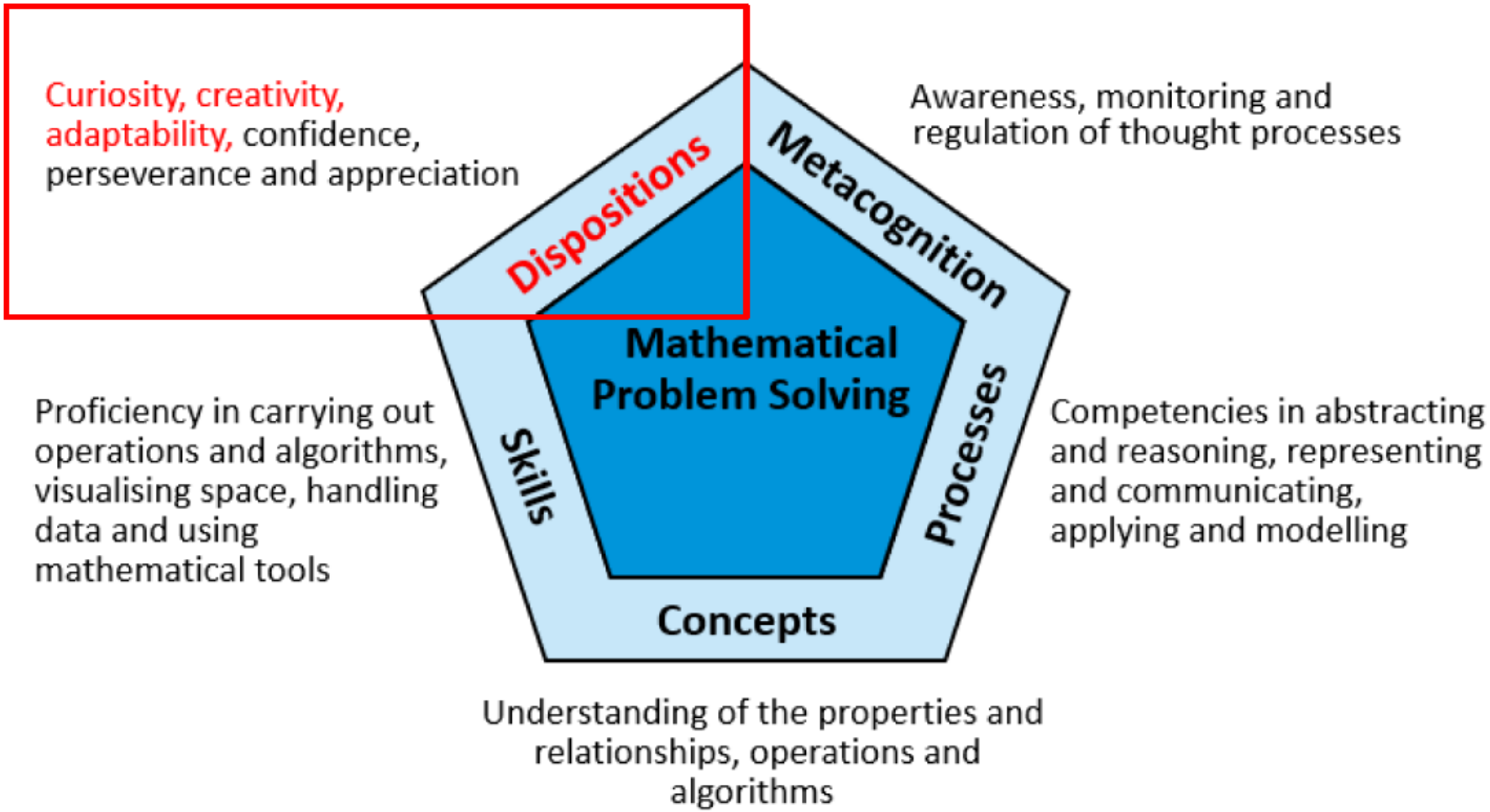


Proficiency in carrying out operations and algorithms, visualising space, handling data and using mathematical tools

Competencies in abstracting and reasoning, representing and communicating, applying and modelling

Understanding of the properties and relationships, operations and algorithms

Singapore Mathematics Framework - The 2028 Pentagon



From Attitudes to Dispositions

Differences and Rationales

- To support 21CC

  (CAIT – Critical, Adaptive, Inventive Thinking)

Belief, appreciation, confidence, motivation, interest and perseverance

Attitudes

- To support STEM

  (3C – Be Curious. Be Creative. Be the Change.)

Curiosity, creativity, adaptability, confidence, perseverance and appreciation

Dispositions

The Framework

- What does it mean to teachers?
- The five (5) components of the framework are integral parts of mathematics learning and problem solving. The intent of the framework is to keep teachers focus on these components in their teaching practice so as to provide more **engaging student-centered** and **technology-enabled** learning environment, and to promote greater **diversity and creativity in learning**.

Syllabus Organisation

Concepts and Skills

Numbers & Algebra	**Geometry & Measurement**	**Statistics & Probability**

Learning Experiences

[Processes, Metacognition and Attitudes/Dispositions]

Example

Content	Learning Experiences
Secondary One	
NUMBER AND ALGEBRA	**Students should have opportunities to:**
N1. Numbers and their operations	
1.1. primes and prime factorisation 1.2. finding highest common factor (HCF) and lowest common multiple (LCM), squares, cubes, square roots and cube roots by prime factorisation 1.3. negative numbers, integers, rational numbers, real numbers and their four operations 1.4. calculations with calculator 1.5. representation and ordering of numbers on the number line 1.6. use of <, >, ≤, ≥ 1.7. approximation and estimation (including rounding off numbers to a required number of decimal places or significant figures, and estimating the results of computation)	(a) **Classify** whole numbers based on their number of factors and **explain** why 0 and 1 are not primes. **[P]** (b) **Make sense of** addition, subtraction and multiplication involving negative integers using algebra discs or other concrete approaches and **develop proficiency** in the 4 operations of integers. **[P]** (c) **Estimate** quantities (numbers and measures) in a variety of contexts, **compare** the estimates and **discuss** the estimation strategies by working in groups. **[P/M/A]** (d) **Compare** follow-through errors arising from intermediate values that are rounded to different degrees of accuracy. **[P]**

Topic	Learning Experiences
Data handling and analysis	***Students should have the opportunities to:*** **Solve** a problem statistically by working collaboratively on a task based on a healthcare or sustainability context that involves: **Posing** a problem and **identifying** the data to be collected **Collecting** data from a representative sample **Classifying** and **representing** data using appropriate statistical representation **Analysing** and **interpreting** data to **make** informed decisions or predictions **Discussing and reflecting** on the ethical uses of data **[CAIT/CCI/DL/STEM/BI/D/M]**

CAIT: Critical, adaptive and inventive thinking		
CCI: Communication, collaboration and information skills		
DL: Digital literacy	**STEM**	
BI: Big ideas	**D:** Dispositions	**M:** Metacognition

Singapore Secondary Mathematics Syllabuses

Section 1: Introduction

Section 2: Mathematics Curriculum

Section 3: Mathematics Syllabuses

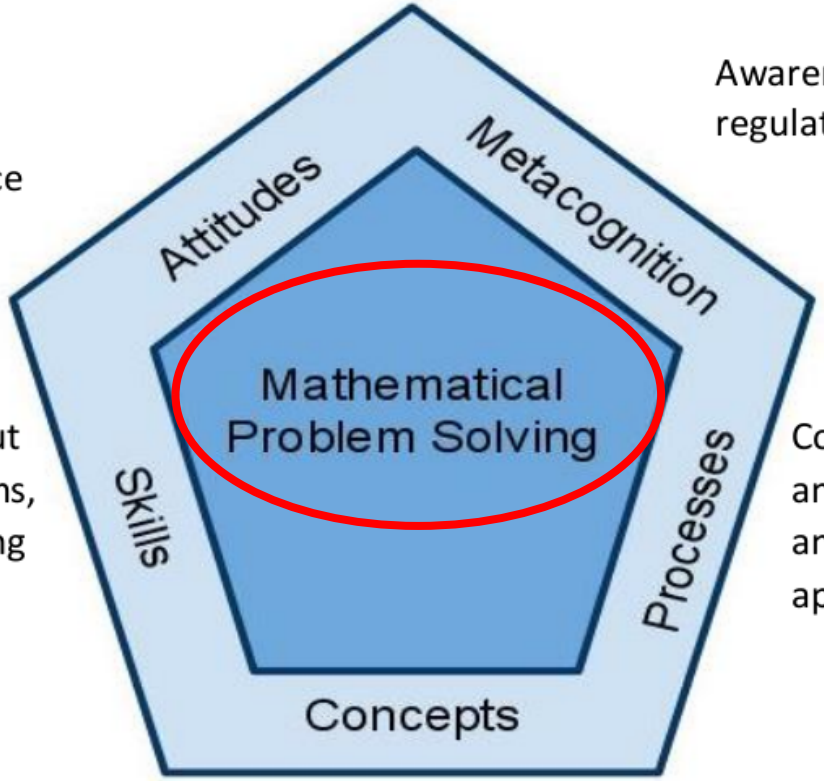
Section 4: Teaching, Learning and Assessing

Section 5: Summative Assessment

Singapore Mathematics Framework - The Current Pentagon

Belief, appreciation, confidence, motivation, interest and perseverance

Awareness, monitoring and regulation of thought processes



Proficiency in carrying out operations and algorithms, visualising space, handling data and using mathematical tools

Competencies in abstracting and reasoning, representing and communicating, applying and modelling

Understanding of the properties and relationships, operations and algorithms

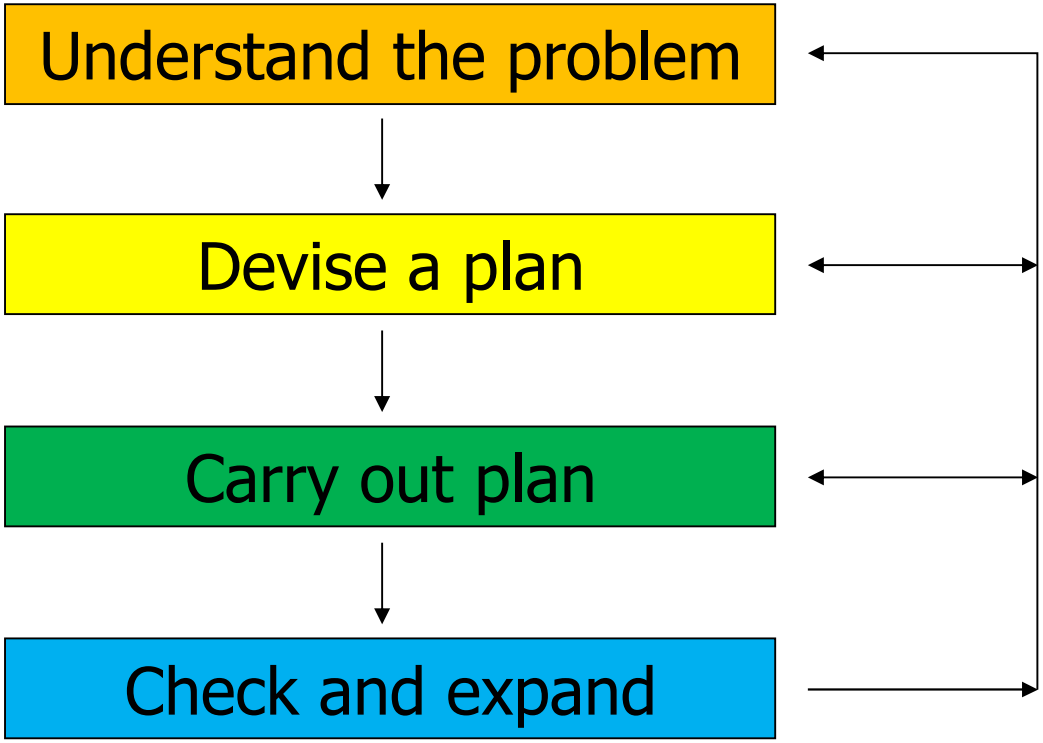
What is a problem?

Mathematical problem solving is a process that takes place when a person who is confronted with a mathematical problem accepts it as a problem to be solved, doesn't have an immediate answer to the problem but makes an attempt to solve it.

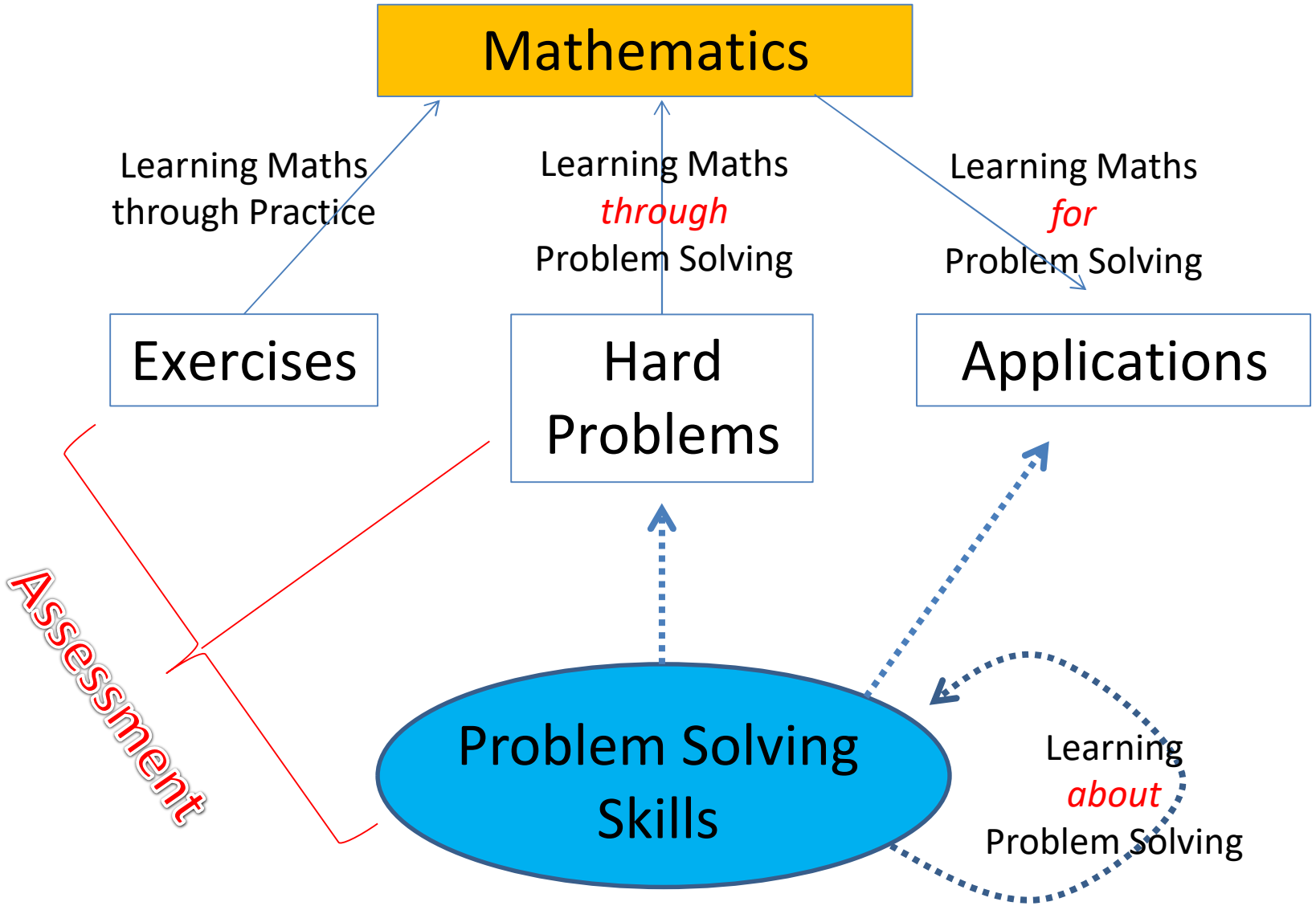
Unpacking

- No easy access to a procedure for solving the problem.
- There must be a desire/motivation/reason to solve the problem
- Problem solving begins when an effort is made to find the solution

Pólya's Problem Solving Strategy



Learning, Mathematics and Problem Solving



References

Hang H.C. (2024). *Singapore Mathematics Curriculum (Secondary School).* PGDE Lecture Notes.

Kaur, B. (2023). *Singapore's Secondary School Mathematics Curriculum*. PGDE Lecture Notes.

Ministry of Education. (2023). *21st Century Competencies*. Received from https://www.moe.gov.sg/education-in-sg/21st-century-competencies on 26 January 2026.

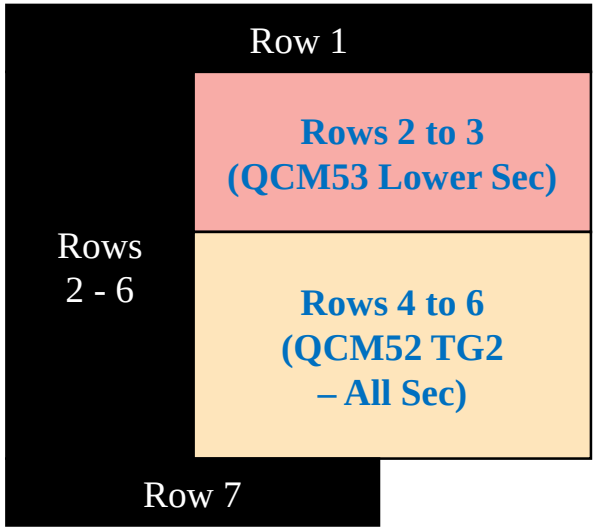
Ministry of Education. (2025). *2020 Secondary Mathematics Syllabuses*.

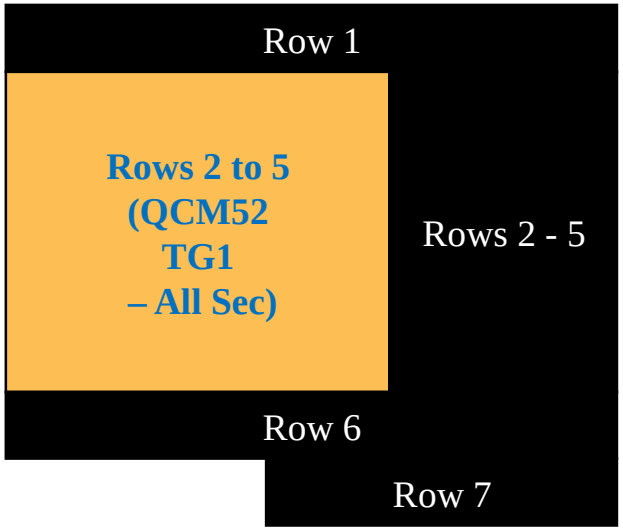
Ministry of Education. (2025). *2025 Secondary Mathematics HOD Seminar* (Slides 17, 29).

Oliva, P.F., & Gordon, W.R.II. (2013). *Developing the Curriculum (8th Ed.)* (pp. 322, 344-346). Pearson Education.

Tay E. G. (2025). The Singapore Mathematics Curriculum, PGDE Lecture Notes.

Screen
(Lecture Table)





Thank you for being early.

Please scan QR or go to Padlet via link below and answer the question in **Section 0.**
What do we teach in Singapore (for the subject mathematics)?

https://tinyurl.com/ssmc2026

Welcome to QCM 52/53.
Please be seated in your designated region.

Not to be seated by students.